

Chapter: Classical Encryption Techniques

1. Playfair Cipher

The Playfair Cipher is a digraph substitution cipher, which means it encrypts two letters (a pair) at a time instead of one letter. It was invented by Charles Wheatstone in 1854, but it became famous through his friend Baron Playfair, so it is called the Playfair Cipher.

Why was it invented?

A simple monoalphabetic cipher (like Caesar Cipher) replaces one letter with another.

Example:

A → D

B → E

C → F

These ciphers are easy to break using frequency analysis because each plaintext letter always maps to the same ciphertext letter. To improve security, Playfair Cipher encrypts pairs of letters, making frequency analysis much harder.

Key Matrix

Playfair Cipher uses a **5 × 5 matrix** of letters.

Since there are 26 English letters, the letters I and J are treated as the same letter.

Steps to create the matrix

Step 1

Choose a keyword.

Example:

MONARCHY

Step 2

Write the letters of the keyword.

Remove repeated letters.

M O N A R C H Y

Step 3

Fill the remaining spaces with the unused letters of the alphabet.

(I and J share one box.)

Final matrix:

M O N A R

C H Y B D

E F G I/J K

L P Q S T

U V W X Z

Encryption Rules

Before encryption:

- Split the plaintext into pairs (digraphs).
- If both letters are the same, insert **X** between them.
- If one letter remains at the end, add **X**.

Example

Plaintext

SECRET MESSAGE

Pairs become

SE CR ET ME SS AG E

Since **SS** contains repeated letters:

SE CR ET ME SX SA GE

Rule 1: Same Row

If both letters are in the **same row**, Replace each letter with the letter **to its right**. If it is the last letter, wrap around to the beginning.

Example:

M O N A R

Encrypt

O → N

R → M

Rule 2: Same Column

If both letters are in the **same column**, Replace each letter with the letter **below it**.

If the letter is at the bottom, wrap around to the top.

Example:

M

C

E

L

U

Encrypt

C → E

U → M

Rule 3: Rectangle Rule

If the two letters are in different rows and columns, Imagine a rectangle. Replace each letter with the letter in the **same row** but in the **column of the other letter**.

Example

M O N A R

C H Y B D

E F G I K

L P Q S T

U V W X Z

Encrypt

M and H

↓

O and C

Example

Plaintext

SECRET MESSAGE

Prepared text

SE CR ET ME SX SA GE

Ciphertext

LI DM KL CL XA XB JF

Decryption

Decryption uses the **opposite** of the encryption rules.

Same Row

Move **left** instead of right.

Same Column

Move **up** instead of down.

Rectangle

Apply the rectangle rule again.

Example

Keyword: MONARCHY

Key Matrix

M O N A R

C H Y B D

E F G I/J K

L P Q S T

U V W X Z

Ciphertext

LI DM KL CL XA XB JF

Pairs:

LI | DM | KL | CL | XA | XB | JF

Decryption Rules

- **Same Row:** Move each letter **left**.
- **Same Column:** Move each letter **up**.
- **Rectangle:** Replace each letter with the letter in the **same row and other letter's column**.

Decryption

Cipher Pair Plain Pair

LI SE

DM CR

KL ET

CL ME

XA SX

XB SA

JF GE

Result:

SE CR ET ME SX SA GE

Remove extra X:

SECRET MESSAGE

Security of the Playfair Cipher

The Playfair cipher is more secure than a monoalphabetic substitution cipher because it encrypts pairs of letters (digrams) instead of single letters.

Key Points

- Encrypts digrams (letter pairs) rather than individual letters.
- Has $26 \times 26 = 676$ possible digrams, making frequency analysis more difficult.
- Requires a 676-entry digram frequency table instead of a 26-letter frequency table.
- Needs more ciphertext for successful cryptanalysis.
- Was widely used by the British and U.S. military during World War I.
- Not completely secure—it can be broken with a few hundred letters of ciphertext because it still preserves some plaintext patterns and language structure.

2. Polyalphabetic Cipher

A Polyalphabetic Cipher uses multiple substitution alphabets instead of only one. Instead of encrypting every letter the same way, different letters may be encrypted using different alphabets. This greatly improves security.

Why use multiple alphabets?

Monoalphabetic Cipher

A always → D

Polyalphabetic Cipher

A → D

A → K

A → M

A → Q

depending on the key.

Therefore, frequency analysis becomes much more difficult.

Features

- Uses multiple substitution alphabets.
- Uses a secret key.
- Repeats the key until the message ends.
- Provides better security than monoalphabetic ciphers.

3. Vigenère Cipher

The Vigenère cipher is a polyalphabetic substitution cipher that encrypts plaintext using a keyword. Unlike the Caesar cipher, it uses multiple substitution alphabets, making it more secure.

How It Works

- A keyword is repeated until it matches the length of the plaintext.
- Each plaintext letter is shifted according to the corresponding letter of the keyword.
- The amount of shift depends on the keyword letter (A = 0, B = 1, C = 2, ..., Z = 25).

Encryption Process

1. Choose a keyword.
2. Repeat the keyword until it matches the plaintext length.
3. Use each key letter as a Caesar Cipher shift.
4. Encrypt every plaintext letter.

- Combine all encrypted letters.

Vigenère Cipher Encryption Example

Plaintext: SECRET MESSAGE

Before encryption, remove the space from the plaintext:

- Plaintext:** SECRETMESSAGE
- Keyword:** LEMON
- Key Stream:** LEMONLEMONLEM

Plaintext	S	E	C	R	E	T	M	E	S	S	A	G	E
Key Stream	L	E	M	O	N	L	E	M	O	N	L	E	M
Shift Value	11	4	12	14	13	11	4	12	14	13	11	4	12
Ciphertext	D	I	O	F	R	E	Q	Q	G	F	L	K	Q

Final Ciphertext:

DIOFREQQGFLKQ

Encryption Process

- Choose the plaintext**
 - Plaintext: **SECRET MESSAGE**
 - Remove spaces → **SECRETMESSAGE**
- Choose a keyword**
 - Keyword: **LEMON**
- Repeat the keyword**
 - Repeat the keyword until it matches the plaintext length:
 - LEMONLEMONLEM**
- Convert letters to numbers**
 - Use the alphabet values:
 - A = 0, B = 1, C = 2, ..., Z = 25
 - Example:
 - S = 18**
 - L = 11**
- Encrypt each letter**
 - Add the plaintext letter value and the keyword letter value.
 - If the result is greater than 25, subtract 26 (or take modulo 26).

Formula:

$$C = (P + K) \bmod 26$$

where:

- C** = Ciphertext letter
- P** = Plaintext letter value
- K** = Keyword letter value

Example Calculations

- S + L**
 - S = 18
 - L = 11
 - $(18 + 11) \bmod 26 = 29 \bmod 26 = 3 = \mathbf{D}$
- E + E**
 - E = 4

- $E = 4$
- $(4 + 4) \bmod 26 = 8 = \mathbf{I}$
- **C + M**
 - $C = 2$
 - $M = 12$
 - $(2 + 12) \bmod 26 = 14 = \mathbf{O}$

Continue the same process for all remaining letters.

6. **Write the encrypted letters together**
 - **Ciphertext: DIOFREQQGFLKQ**

Decryption Process

1. Write the repeated keyword.
2. Use the key letter to reverse the Caesar shift.
3. Recover the original plaintext.

Vigenère Cipher Decryption

Keyword: LEMON

Ciphertext: DIOFREQQGFLKQ

Repeat the keyword until it matches the ciphertext length:

- **Ciphertext:** DIOFREQQGFLKQ
- **Key Stream:** LEMONLEMONLEM

Ciphertext	D	I	O	F	R	E	Q	Q	G	F	L	K	Q
Key Stream	L	E	M	O	N	L	E	M	O	N	L	E	M
Shift Value	11	4	12	14	13	11	4	12	14	13	11	4	12
Plaintext	S	E	C	R	E	T	M	E	S	S	A	G	E

Final Plaintext:

SECRET MESSAGE

Decryption Process

1. **Take the ciphertext**
 - Ciphertext: **DIOFREQQGFLKQ**
2. **Repeat the keyword**
 - Keyword: **LEMON**
 - Key Stream: **LEMONLEMONLEM**
3. **Convert letters to numbers**
 - Use the alphabet values:
 - $A = 0, B = 1, C = 2, \dots, Z = 25$
4. **Decrypt each letter**
 - Subtract the keyword letter value from the ciphertext letter value.
 - If the result is negative, add **26**.

Formula:

$$P = (C - K + 26) \bmod 26$$

where:

- **P** = Plaintext letter
- **C** = Ciphertext letter value
- **K** = Keyword letter value

Example Calculations

- **D – L**
 - $D = 3$
 - $L = 11$
 - $(3 - 11 + 26) \bmod 26 = 18 = \mathbf{S}$
- **I – E**
 - $I = 8$
 - $E = 4$
 - $(8 - 4) \bmod 26 = 4 = \mathbf{E}$
- **O – M**
 - $O = 14$
 - $M = 12$
 - $(14 - 12) \bmod 26 = 2 = \mathbf{C}$

Continue the same process for all remaining letters.

Advantages

- Much stronger than Caesar Cipher.
- Much stronger than Playfair Cipher for short messages.
- Hides normal letter frequencies.

4. Security of Vigenère Cipher

The Vigenère Cipher uses multiple alphabets. As a result, the same plaintext letter may become different ciphertext letters.

Example

$A \rightarrow D$

$A \rightarrow K$

$A \rightarrow P$

This hides frequency patterns.

However, the frequencies are **not completely hidden**.

With enough ciphertext, the key length can often be discovered.

5. Kasiski Method

The Kasiski Method (or Kasiski Examination) is a cryptanalysis technique used to break the Vigenère cipher. It works by finding repeated sequences of letters in the ciphertext and measuring the distance between them to estimate the length of the secret keyword.

Basic Idea

If the same word appears several times in the plaintext, and both occurrences are encrypted using the same part of the key, they produce the same ciphertext. Repeated ciphertext patterns reveal the key length.

Steps

Step 1

Look for repeated groups of letters in the ciphertext.

Example

VTW

appears more than once.

Step 2

Measure the distance between repeated groups.

Suppose

VTW

appears every

9 letters

Step 3

Find the factors of the distance.

Factors of 9 are

1

3

9

The key length is likely

3

or

9

Step 4

Separate the ciphertext into groups based on the key length. Each group behaves like a monoalphabetic cipher.

Step 5

Use frequency analysis on each group. Eventually, the original keyword can be recovered.

Example

Ciphertext

ZICVTWQNGRZGVTWAVZHCQYGLMGJ

Repeated part

VTW

This suggests a possible key length of

3

or

9

Then each alphabet is attacked separately.

Advantages of Kasiski Method

- Finds the probable key length.
- Makes breaking the Vigenère Cipher much easier.
- Works well when enough ciphertext is available.